

Abdoulaye Zakaria Djerou

Software Developer · MSIT Student at Carnegie Mellon University Africa · Full-Stack & Backend Engineering

Kigali, Rwanda · azdjerou@gmail.com · +250 791 375 009 · github.com/kardara · kardara.vercel.app · LinkedIn

PROFILE

Software Developer and Master of Science in Information Technology student at Carnegie Mellon University Africa, specializing in Artificial Intelligence and Machine Learning. Experienced in building full-stack and backend applications with Java, Spring Boot, React, Next.js, TypeScript and PostgreSQL: academic information systems, REST APIs, authentication and authorization, database design, and deployment on Linux servers.

Alongside software development, experienced in teaching, technical training, community leadership and mentoring. Particularly interested in software engineering, artificial intelligence, research, and using technology to address challenges in Africa.

EDUCATION

Master of Science in Information Technology (MSIT) — Carnegie Mellon University Africa, Kigali Specialization: Artificial Intelligence & Machine Learning	2026 – Present
Bachelor's Degree in Information Technology — Adventist University of Central Africa (AUCA), Kigali	2023 – 2026
Baccalaureate (Series D — Science) — Lycée Ibnou Cina, Chad	2022

PROFESSIONAL EXPERIENCE

Software Developer <i>Adventist University of Central Africa (AUCA), Kigali</i> - Develop and maintain full-stack solutions for university information systems. - Build backend services with Java, Spring Boot and PostgreSQL, and frontends with React, Next.js and TypeScript. - Design and integrate REST APIs and database-driven applications. - Contribute to system architecture, debugging, deployment and continuous improvement. - Work with stakeholders to translate requirements into practical software solutions.	2026 – Present
Lead Software Engineer <i>ChadNova, Kigali</i> - Led software development initiatives and contributed to technical decision-making. - Designed and developed full-stack applications with Java, Spring Boot, React, Next.js, TypeScript and PostgreSQL. - Contributed to software architecture, database design, deployment and system maintenance. - Collaborated with the team on debugging, testing and continuous improvement of maintainable, scalable systems.	Nov 2025 – Aug 2026
Teaching Assistant — Web Technology & Internet <i>Adventist University of Central Africa (AUCA), Kigali</i> - Supported practical sessions on React, Java Servlets, Spring Boot, JPA, REST APIs and web technologies. - Helped students with programming exercises, technical troubleshooting, and frontend and backend projects. - Assisted with assignments, examinations, grading and academic activities. - Worked with React, Tailwind CSS, Vite, Java, Spring Boot, PostgreSQL, Axios and the Fetch API.	Sep 2025 – Apr 2026
Advanced Java Backend & Software Architecture Training <i>The Gym x MaibornWolff, Kigali</i> Advanced Java · Object-Oriented Programming · Spring Boot · PostgreSQL · Software Architecture · Multithreading & Concurrency · Algorithms & Problem Solving · Backend System Design · Clean Code	Feb 2026 – Present
Volunteer Coach <i>The Gym, Kigali</i> - Supported beginner members during training sessions on proper technique and consistent training habits. - Contributed to an inclusive, supportive environment; developed communication, leadership and mentoring skills.	Feb 2026 – Jun 2026

SELECTED PROJECTS

AUCA Integrated Management System (AUCA IMS) — Full-Stack Software Developer Stack: Next.js · React · TypeScript · Spring Boot 3.5 · Java 21 · PostgreSQL · jOOQ · Flyway · Spring Security · JWT · Docker · Linux A modern academic management platform replacing legacy registration processes with a centralized digital system. - Designed and developed a modular full-stack system: Next.js/React/TypeScript frontend and Spring Boot/Java 21 backend. - Designed a PostgreSQL database with 40+ tables and triggers; migrations with Flyway and type-safe access with jOOQ. - Implemented JWT authentication in HttpOnly cookies and a dynamic RBAC system with 50+ permissions across Student, Admin, Registrar, Lecturer, HOD, Dean and Finance roles.	github.com/kardara/auca-ims-frontend
---	--------------------------------------

- Implemented a multi-stage registration workflow: PENDING → STUDENT_VALIDATED → DEPARTMENT_VALIDATED → REGISTRAR_VALIDATED → FINANCE_VALIDATED.
- Contributed to Docker-based deployment and Linux server administration; designed for ~1,500–2,000 concurrent registration users.

ChadNova CleanEX — Software Developer

A service initiative creating cleaning opportunities and regular work for vulnerable community members. Contributed to the development and technical implementation of the platform supporting service coordination and management. The initiative currently supports 18 widows with regular work.

StudyBuddy — Full-Stack Developer

Stack: ASP.NET · C# · OpenAI · OpenRouter

An educational platform supporting students with learning and academic assistance. Developed the ASP.NET backend, integrated AI services through OpenAI and OpenRouter, and worked across application logic, APIs and user-facing features.

Web Technology & React Projects — Developer / Teaching Assistant

Stack: React · JavaScript · Vite · Tailwind CSS · Axios · REST APIs

Practical teaching projects covering React fundamentals, state and forms, CRUD applications, API integration, React Router, authentication concepts and component-based design.

TECHNICAL SKILLS

Languages	Java, JavaScript, TypeScript, Python, C#, SQL, HTML, CSS
Frontend	React.js, Next.js, Tailwind CSS, Vite, Axios, React Hooks, React Router, Responsive Web Development
Backend	Spring Boot, Spring Web, Spring Security, Spring Data JPA, Hibernate, Java Servlets, REST APIs, Node.js, Express.js, ASP.NET
Databases	PostgreSQL, MySQL, MongoDB, Firebase, Supabase
DevOps & Tools	Git, GitHub, GitHub Actions, Docker, Linux, Maven, Flyway, jOOQ, JWT, Netlify, Vercel
Engineering	OOP, RESTful API Design, Database Design, Software Architecture, Authentication & Authorization, RBAC, System Design, Testing, Debugging

LEADERSHIP & COMMUNITY ENGAGEMENT

Secretary-General & Spokesperson — Independent Electoral Commission, Association of Chadian Students and Trainees in Rwanda (AEESTR) 2025
Organized and coordinated the association's electoral process and managed its communication; later served as an advisor within the association.

Secretary-General — Beri Bour Rwanda
Coordinated community activities, helped organize the annual Beri Bour Cultural Day, and supported cultural preservation initiatives.

Treasurer — AC-DEV Sec-Rwanda
Supported financial organization, administration, and the planning of community initiatives.

Lead Organizer — Chadian Independence Day 2025

Member — Toastmasters International

TRAINING & CERTIFICATIONS

Full Stack Development Training — React.js and Spring Boot	Online	2024 – 2025
Cisco Networking Essentials	Cisco Networking Academy (Online)	2024
Graphic Design (Adobe Photoshop)	Online	2023 – 2024
Nursing Internship	Hôpital Militaire de Garnison, N'Djamena	Aug – Oct 2022
Leadership & Team Management	Local Youth Program	2021
Professional Nursing Program	EFATA School, Sarh	Aug – Sep 2020
Red Cross Humanitarian Training Program	Croix-Rouge, N'Djamena	2020

LANGUAGES & INTERESTS

Languages: French, English, Arabic (Fluent) · Spanish (Beginner)

Interests: Software Engineering, AI & Machine Learning, Research, Science & Technology, Technology for Africa, Education, Mentoring, Community Engagement, Fitness